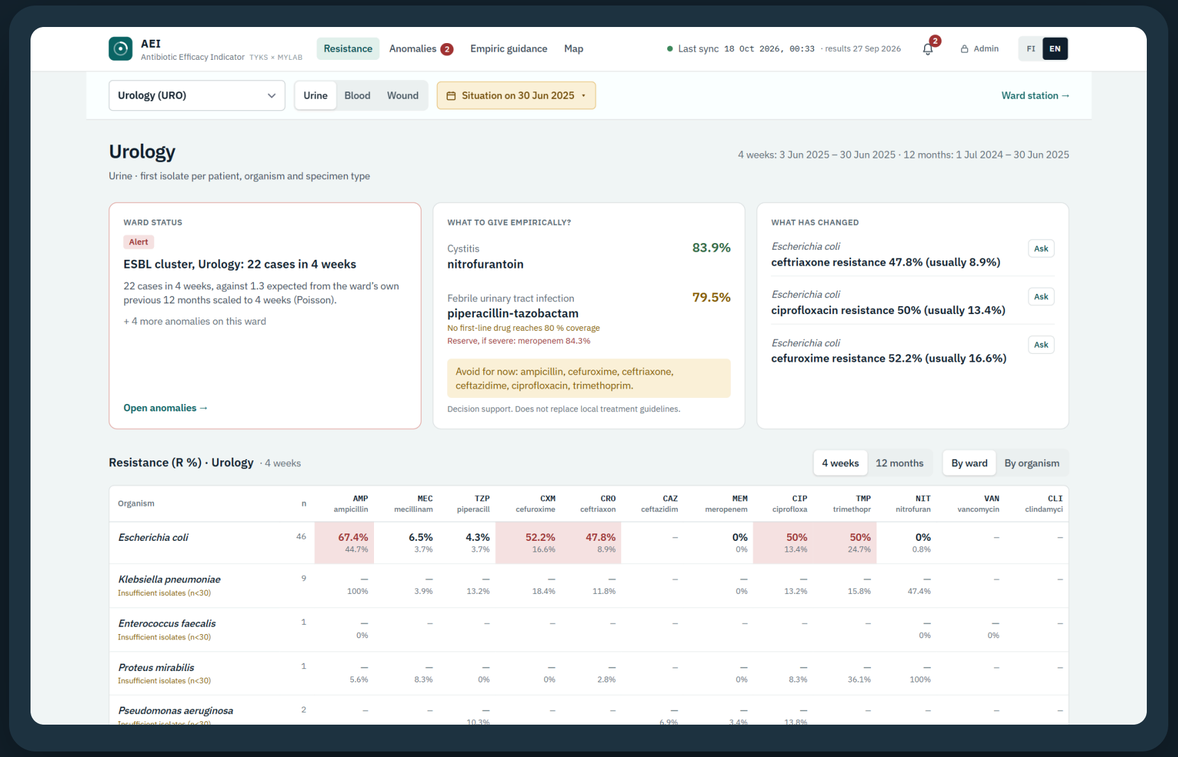
TYKS LABORATORIES × MYLAB

AEI

Antibiotic Efficacy Indicator

Which antibiotics still work — ward by ward, this month. A calm surveillance layer on top of the laboratory's results.

JunctionX Turku 2026 · Finnish-first · English · local AI



THE PROBLEM

Resistance changes faster than the reports that describe it

TODAY · 1

Yearly, hospital-wide antibiograms

A shift in one ward stays invisible until the next annual summary.

TODAY · 2

Clusters spotted by people

Infection control reads result lists by hand; an outbreak shows when cases have piled up.

TODAY · 3

Empiric therapy from general tables

Guidance reflects the country or the hospital, not this ward’s last four weeks.

AEI turns the same laboratory results into a rolling, ward-level signal — without touching any clinical system.

HOW IT WORKS

A shadow layer: it reads, computes and flags — nothing else

01
Lab extract
Cultures, isolates, S/R results · read-only

02
Canonical schema
WHONET codes, EUCAST S/R, validated

03
Rolling windows
First isolates · 4 weeks vs 12 months

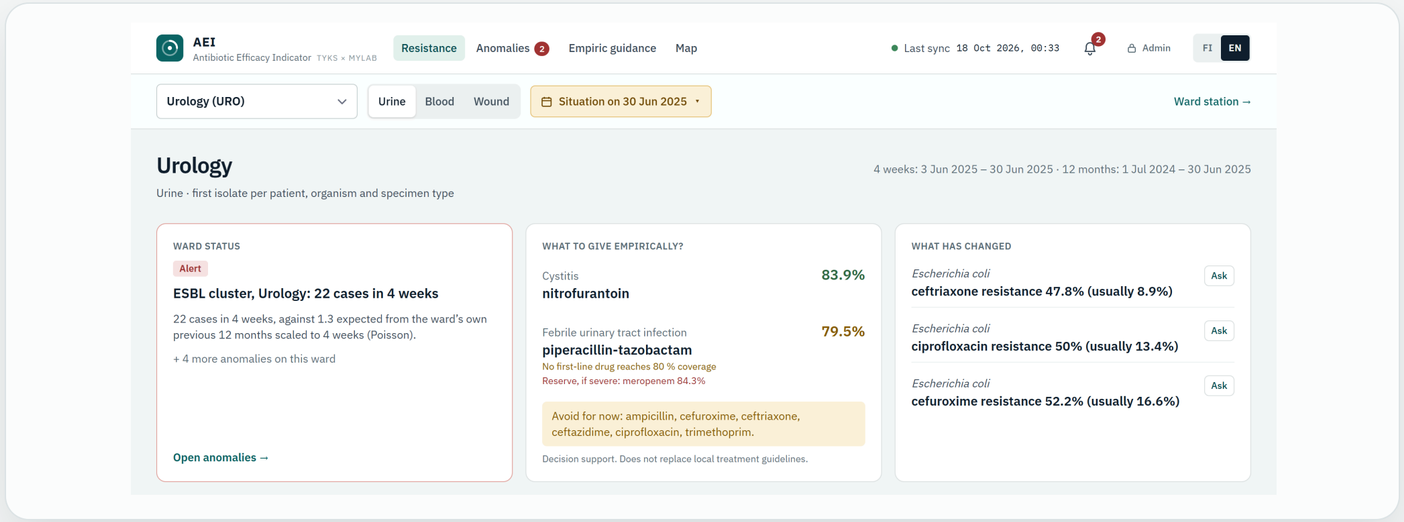
04
System-1 triage
Clusters, sentinels, rises, surges

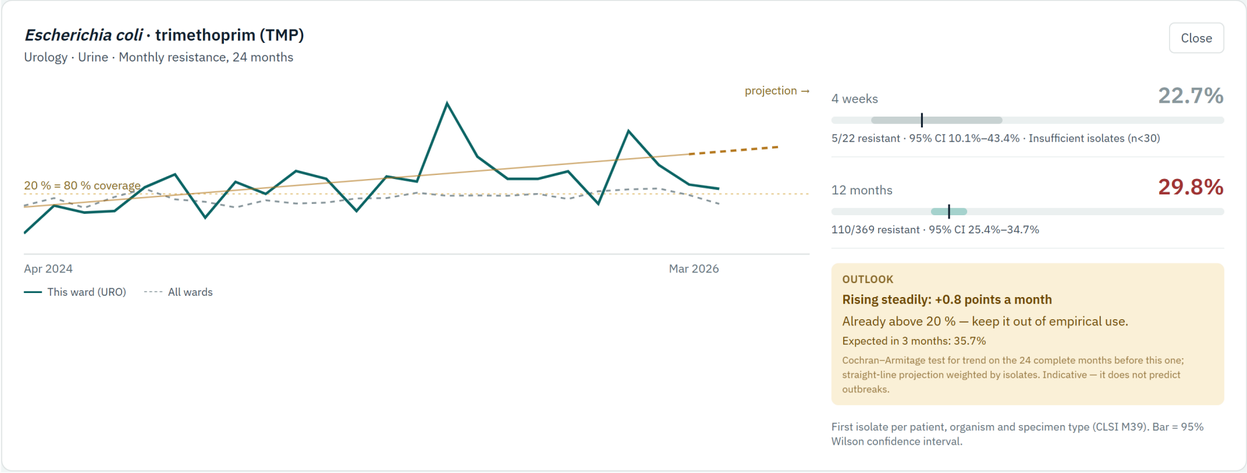
05
Ward views
Resistance, alerts, guidance, chat

Every number from code
The AI words answers; it never computes them.

Each ward vs itself
Baselines are the ward’s own previous year.

No % below 30 isolates
CLSI M39 first-isolate rule, Wilson intervals.



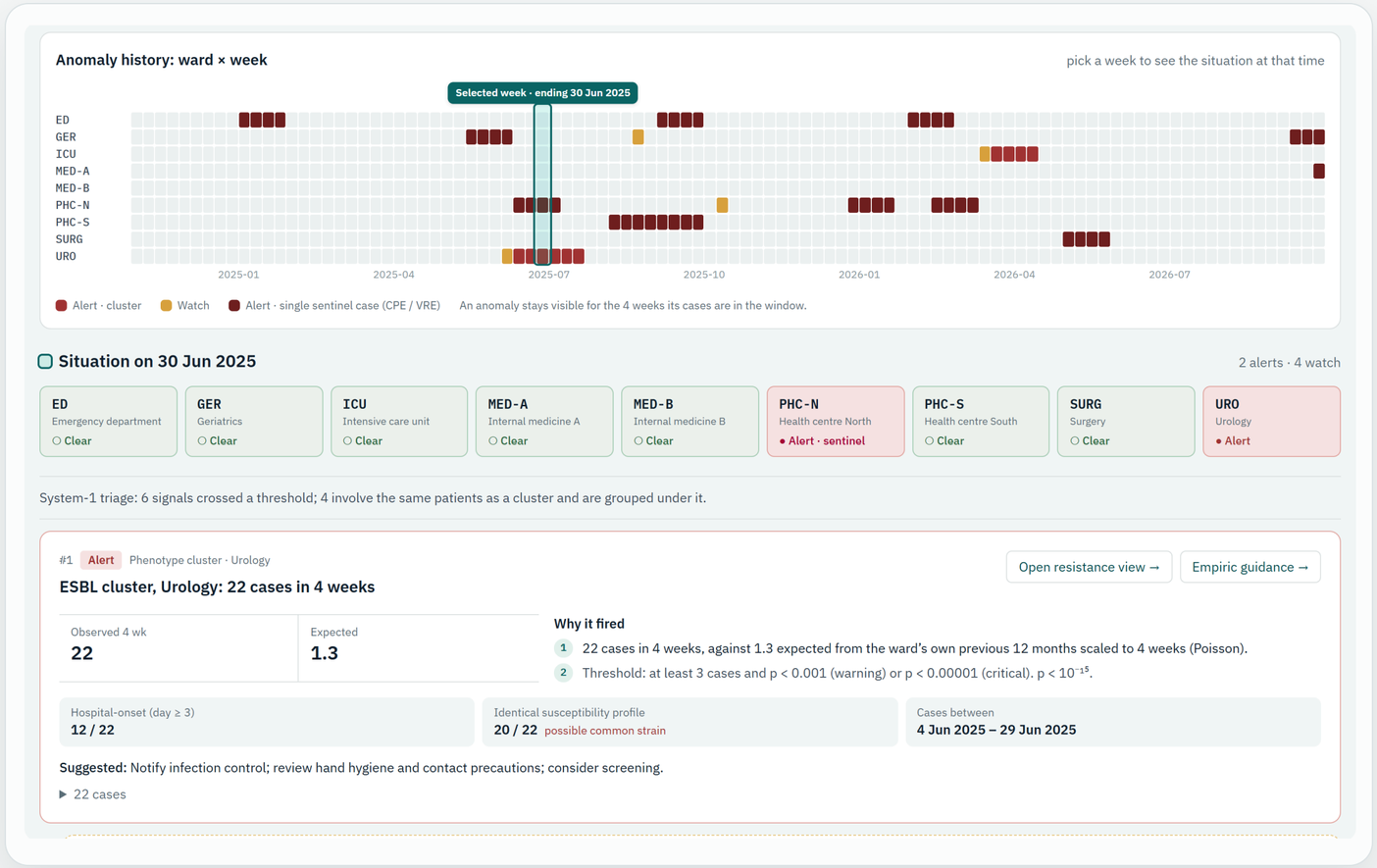


01 · RESISTANCE

One page per ward, answer first

- **Three filters** — ward, specimen, date
- **Three answers** — status, what to give, what changed
- ↗ **Early warning** — slow drifts, projected to the 20 % line

Added value: one calm view that warns before a drug stops working — not after.

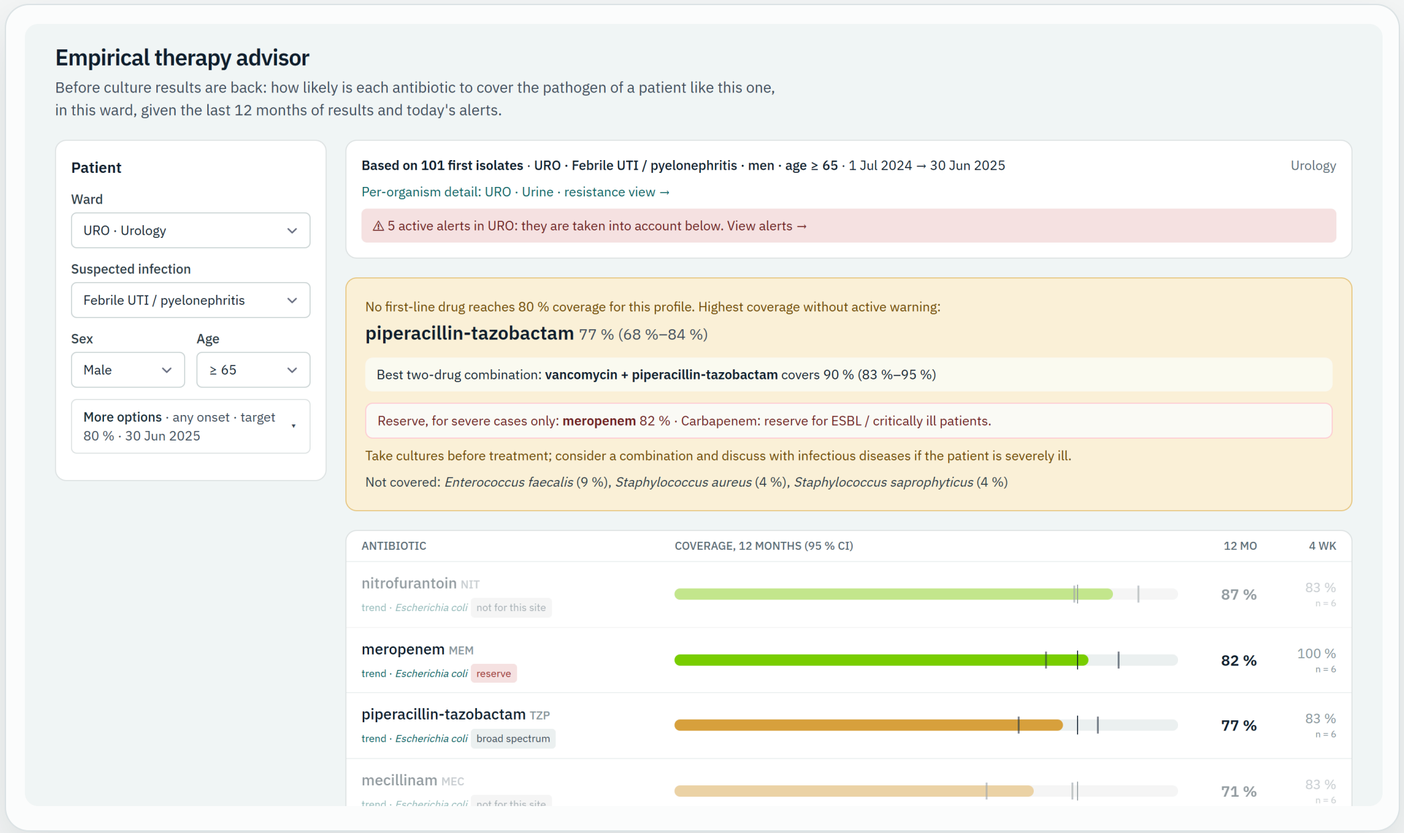


02 · ANOMALIES

Calm detection that explains itself

- **Detectors** — ESBL/MRSA clusters, CPE/VRE sentinels, resistance rises, surges
- **Why it fired** — observed vs expected, rule, p-value
- **Triage** — 6 signals, 2 events: rises with the same patients sit under their cluster

Added value: the seeded ESBL outbreak is flagged in its first week — 22 cases vs 1.3 expected — with no list to read.

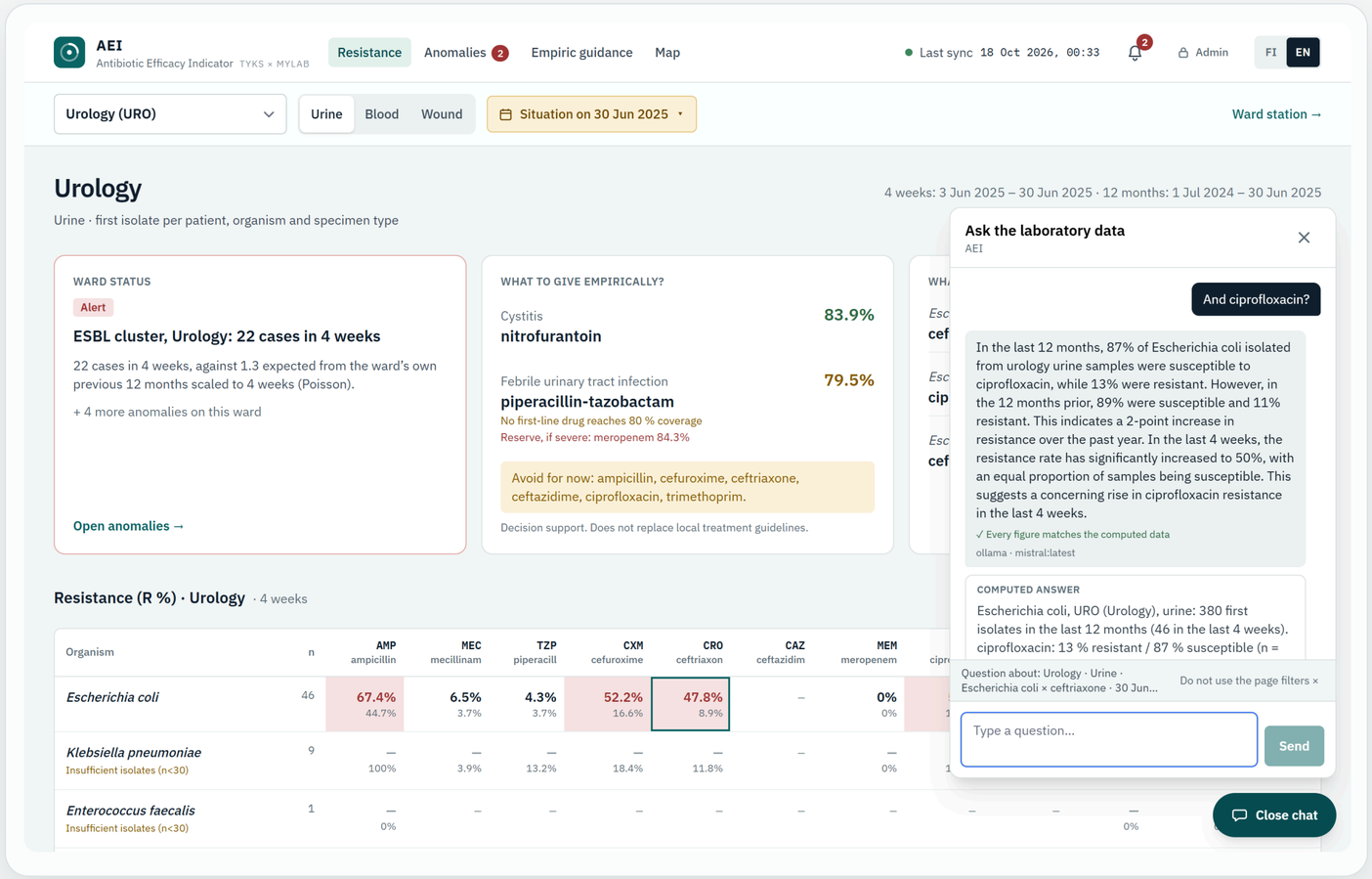


03 · EMPIRIC GUIDANCE

What to give before the culture is back

- **Patients like this one** — ward, infection, sex, age, onset
- **Narrowest first-line drug** — carbapenems kept in reserve
- **Alert-aware** — drugs weakened by an active cluster are flagged

Added value: local, current and stewardship-minded — the reserve drug is never the first suggestion.



04 · ASK AEI

Ask in Finnish, English or French

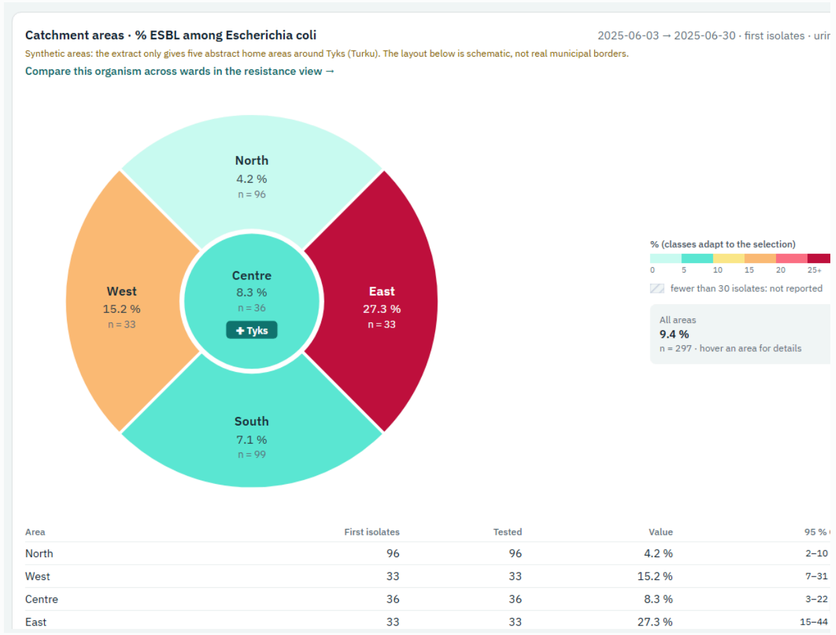
- **Local model** — mistral via Ollama, inside the hospital
- **Code computes, AI words** — every figure is checked
- **Knows the page** — ward, specimen, the opened cell: “And ciprofloxacin?” is enough

Added value: a natural-language door to the data, with no patient row ever sent to the model.

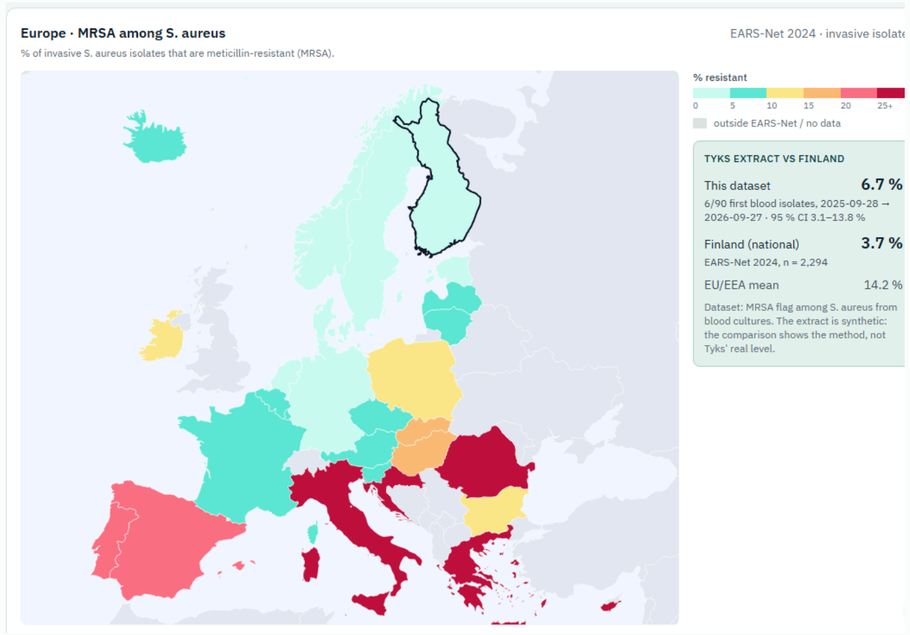
05 · MAP

Local hotspots, national benchmark

Added value: where resistance comes from, and how Tyks compares with Finland and Europe.



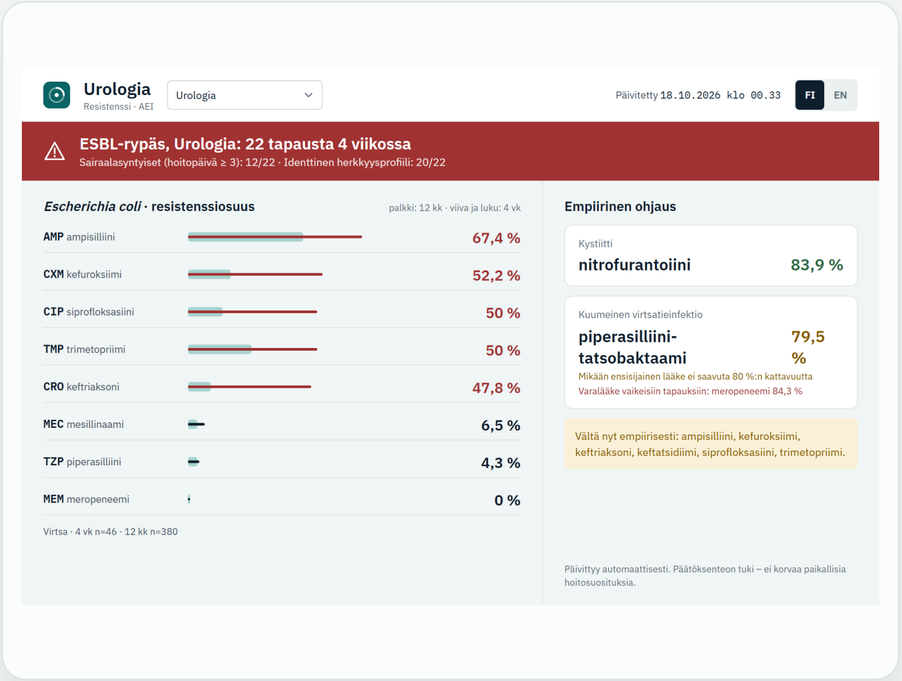
Home areas — % resistant, ESBL or MRSA by catchment; East stands out at 27 % ESBL



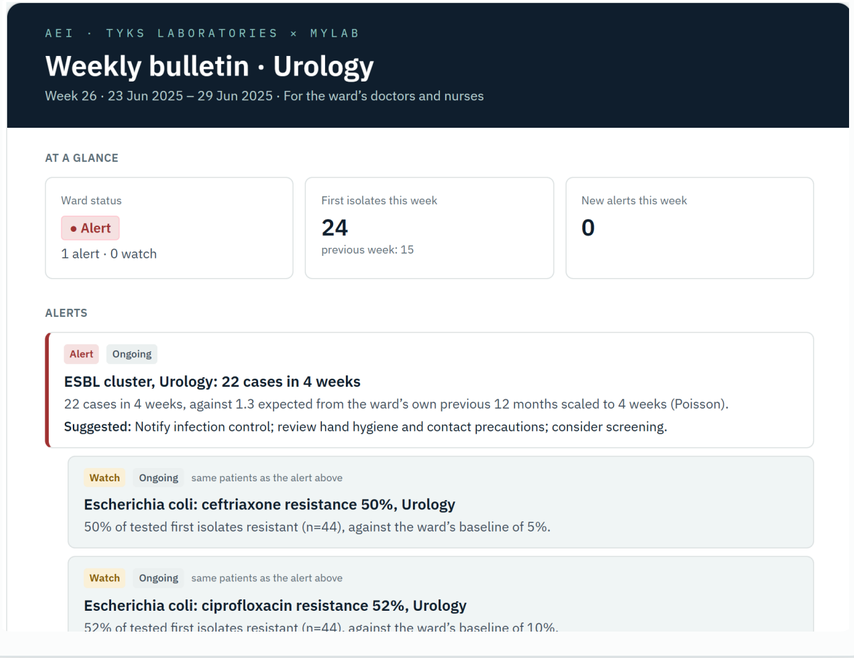
Europe — real ECDC EARS-Net 2024 figures, 29 countries; Finland highlighted

On the ward, every Monday, on a phone

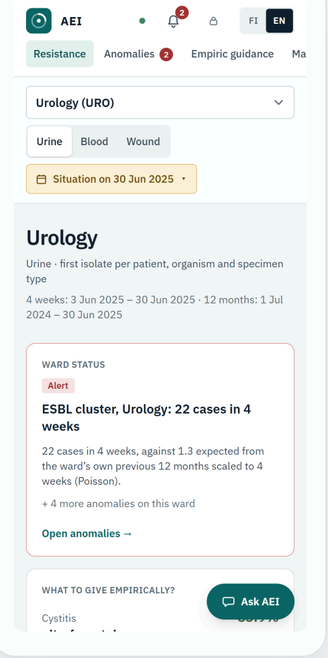
Added value: AEI comes to the ward — a corridor screen, a Monday bulletin, a phone.



Ward station — a corridor screen, Finnish-first, refreshes itself



Weekly bulletin — every Monday, one per ward and one for infection control, printable



Mobile — same answers, full-screen chat

ADDED VALUE

From yearly reports to a ward-level signal

Need	Typical today	With AEI
Freshness	Annual summaries	Rolling 4 weeks vs 12 months, reloaded within 30 s
Granularity	Hospital-wide	Ward × organism × drug, n ≥ 30 rule
Outbreaks	Spotted by reading lists	Automatic triage that explains why it fired
Empiric choice	General guidance	Local coverage that reacts to active alerts
Access	Specialist reports	Finnish-first views, ward screen, chat in 3 languages
Anticipation	Seen afterwards	Slow drifts flagged and projected to the 20 % line
Proactivity	You go and look	A bulletin per ward every Monday, printable
Data safety	—	Read-only, local AI, aggregates only

Kiitos!

Next: live Mylab LIS feed · acknowledgement workflow with audit trail · Finnish on every page